

Underwater Vision-Based Surveillance Using Machine Learning and Artificial Intelligence

DAY 2 — Literature Review: Domain + ML

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1. Objective

The objective of Day 2 is to study existing research related to Computer Vision, Underwater Imaging, Deep Learning, and Visual Surveillance.

The literature review focuses on understanding existing underwater image enhancement methods, object detection methods, underwater datasets, and tracking approaches. The reviewed works are compared based on their data/datasets, approaches, evaluation metrics, results, and limitations.

Based on this review, suitable candidate methods are shortlisted for further experimentation on all three supplied datasets.

2. Core Domain Concepts

2.1 Computer Vision

Computer Vision is a field of Artificial Intelligence that enables computers to understand and analyze images and videos.

In this project, Computer Vision is used to identify and analyze underwater objects from images or video.

2.2 Underwater Imaging

Underwater images often suffer from:

- Color distortion
- Low contrast
- Blur
- Noise
- Poor visibility
- Uneven illumination

These problems can make object detection more difficult. Therefore, image enhancement can be considered before object detection.

2.3 Deep Learning

Deep Learning uses neural networks to automatically learn patterns from large amounts of data.

In this project, deep learning can be used for object detection, where the model identifies objects and their locations in underwater images.

2.4 Visual Surveillance

Visual Surveillance involves analyzing images or video continuously to detect and monitor objects or activities.

For this project, object detection can identify underwater objects, while tracking can be used to follow detected objects across video frames when sequential/video data is available.

3. Supplied Datasets

The project provides three datasets.

Dataset	Number of Classes	Classes
well.v8i.yolov8	4	Inlet-pipe, fishes, school-of-fish, stone
DIATAquarium.v4i.yolov8	11	Aquatic Plant, Camera, Conch-Shell, Fish, Fishes, Pluco, Shark, Temperature Sensor, Water-Pump, Water-filter, betta
Aquatic Plant.v2i.yolov8	1	Aquatic_plant

All three datasets will be considered in the project as required. Since their classes are different, they will initially be treated as separate datasets during evaluation.

4. Literature Review

Paper 1 — Er et al. (2023)

Title: *Research Challenges, Recent Advances, and Popular Datasets in Deep Learning-Based Underwater Marine Object Detection: A Review*

Data / Dataset

The paper reviews commonly used underwater object-detection datasets and existing research.

Approach

It provides a review of deep-learning-based underwater marine object detection methods.

Metrics

- Precision
- Recall
- mAP
- Detection speed

Results / Findings

- Poor underwater image quality
- Small objects
- Difficulty generalizing across environments

- Requirement for real-time detection

Limitations

As a review paper, it does not propose one new detection model with a single experimental result.

Why relevant?

It gives an overall understanding of the underwater object-detection research area and its challenges.

Paper 2 — Xu et al. (2023)

Title: A Systematic Review and Analysis of Deep Learning-Based Underwater Object Detection

Data / Dataset

The study reviews existing underwater object-detection research and the datasets used by those studies.

Approach

It systematically analyzes deep-learning-based underwater object detection and the relationship between image enhancement and detection.

Metrics

- Precision
- Recall
- AP
- mAP

Results / Findings

An important finding is that better-looking enhanced images do not necessarily produce better object-detection accuracy. Image enhancement can sometimes introduce changes that create a domain shift for the detector.

Limitations

The study is a systematic review rather than a single new detection system.

Why relevant?

This is particularly relevant because our project may compare original images versus enhanced images for object detection.

Paper 3 — Li et al. (2019)

Title: An Underwater Image Enhancement Benchmark Dataset and Beyond

Data / Dataset

The paper introduces UIEB (Underwater Image Enhancement Benchmark Dataset) containing 950 real-world underwater images. Among them, 890 have reference images and 60 challenging images do not have satisfactory reference images.

Approach

The paper proposes Water-Net, a learning-based underwater image enhancement method.

Metrics

The work uses quantitative and qualitative evaluation of enhanced images. Common image-quality measures include:

- PSNR
- SSIM
- UIQM

Results / Findings

The study provides a benchmark for evaluating underwater image enhancement and demonstrates the usefulness of Water-Net for improving underwater image quality.

Limitations

The main focus is image enhancement, not underwater object detection.

Why relevant?

It provides a well-known underwater enhancement dataset and a candidate learning-based enhancement method.

Paper 4 — Islam, Xia & Sattar (2020)

Title: Fast Underwater Image Enhancement for Improved Visual Perception

Data / Dataset

The work evaluates underwater image enhancement using underwater image datasets such as UIEB/EUVP-related data.

Approach

The paper proposes FUnIE-GAN, a fast learning-based underwater image enhancement method.

Metrics

The work considers image-quality measures such as SSIM, PSNR, and UIQM. It also investigates the usefulness of enhancement for downstream vision tasks.

Results / Findings

FUnIE-GAN is designed to provide underwater enhancement while being computationally efficient enough for fast visual perception.

Limitations

It focuses mainly on enhancement and visual perception rather than being a complete underwater surveillance/detection system.

Why relevant?

FUnIE-GAN is a candidate learning-based enhancement method for our project.

Paper 5 — Zhang et al. (2023)

Title: *An Improved YOLOv5-Based Underwater Object-Detection Framework*

Data / Dataset

- URPC2019
- URPC2020

Approach

The paper proposes an improved YOLOv5-based underwater object-detection framework using improvements to the network architecture and enhancement-related techniques.

Metrics

- AP
- mAP@0.5
- mAP@0.5:0.95

Results

- 79.8% mAP@0.5 on URPC2019
- 79.4% mAP@0.5 on URPC2020
- 44.2% mAP@0.5:0.95 on URPC2019

Limitations

The method is more complex than a basic YOLO implementation and was evaluated on specific underwater datasets.

Why relevant?

It demonstrates the use of a YOLO-based detector for underwater object detection, which is directly relevant to our detection stage.

Paper 6 — Liu et al. (2021)

Title: *A Dataset and Benchmark of Underwater Object Detection for Robot Picking*

Data / Dataset

The paper introduces the DUO dataset for underwater object detection.

Approach

It establishes a benchmark for evaluating underwater object-detection systems, with attention to both detection accuracy and efficiency.

Metrics

- Detection accuracy
- Detection efficiency/speed

Results / Findings

- Similar-looking underwater objects
- Incomplete annotations
- Lack of suitable test annotations in previous datasets

Limitations

The dataset and benchmark are designed around underwater robot-picking scenarios, so the environment and object categories may differ from our datasets.

Why relevant?

It helps us understand how underwater detection datasets and benchmarks are designed and evaluated.

Paper 7 — Fu et al. (2023)

Title: Rethinking General Underwater Object Detection: Datasets, Challenges, and Solutions

Data / Dataset

The paper introduces the RUOD dataset, containing approximately 14,000 images, 74,903 labeled objects, and 10 categories.

Approach

The study investigates general underwater object detection and proposes solutions for dealing with underwater environmental variations.

Metrics

- Precision
- Recall
- AP
- mAP

Results / Findings

- Haze-like effects
- Color cast
- Light interference
- Different underwater environments

Limitations

The dataset and proposed solutions may not directly match the classes and characteristics of our three supplied datasets.

Why relevant?

It is useful for understanding general underwater object detection and dataset/environment variation.

5. Paper / Method Comparison Table

This is the main table required by the Day 2 instructions.

Paper / Method	Data / Dataset	Approach	Metrics	Results / Findings	Limitations
Er et al. (2023)	Underwater datasets reviewed	Review of deep-learning underwater detection	Precision, Recall, mAP, speed	Identifies image degradation, small objects, generalization and real-time detection as major challenges	Review paper; no single proposed detector
Xu et al. (2023)	Underwater detection studies/datasets	Systematic review of underwater detection and enhancement	Precision, Recall, AP, mAP	Enhancement does not always improve detection; domain shift can occur	Review rather than a new detection model
Li et al. (2019)	UIEB — 950 images	Water-Net enhancement	PSNR, SSIM, UIQM and qualitative evaluation	Provides benchmark for underwater image enhancement	Focuses on enhancement rather than detection
Islam et al. (2020)	Underwater enhancement datasets	FUnIE-GAN	SSIM, PSNR, UIQM and downstream evaluation	Fast learning-based underwater image enhancement	Mainly focused on enhancement/visual perception
Zhang et al. (2023)	URPC2019, URPC2020	Improved YOLOv5-based detector	AP, mAP@0.5, mAP@0.5:0.95	79.8% mAP@0.5 on URPC2019; 79.4% on URPC2020	More complex model; evaluated on specific datasets
Liu et al. (2021)	DUO underwater dataset	Underwater detection benchmark	Accuracy and efficiency	Addresses dataset/annotation and benchmark problems	Robot-picking scenario may differ from our project
Fu et al. (2023)	RUOD — 14,000 images, 74,903 objects	General underwater object detection	Precision, Recall, AP, mAP	Highlights environmental variation and general underwater detection challenges	Dataset/classes differ from our supplied datasets

6. Candidate Methods

Based on the literature review, the following methods are shortlisted as candidate methods for the next stages.

A. Classical Image Enhancement

White Balance

- Attempts to correct underwater color distortion.

CLAHE

- Improves local image contrast.
- Can help reveal details in low-contrast underwater images.

These are simple classical baselines.

B. Learning-Based Image Enhancement

Water-Net

- Learning-based underwater image enhancement method.
- Selected from the literature as a candidate enhancement approach.

FUnIE-GAN

- Fast learning-based underwater image enhancement method.
- Selected as another candidate enhancement approach.

C. Transfer-Learning-Based Object Detector

YOLO-based object detector

A YOLO detector will be considered as the candidate deep-learning object detector.

Transfer learning can be used by starting from a pretrained model and fine-tuning it on the supplied underwater datasets.

The detector will be considered for all three supplied datasets.

D. Simple Tracker

A simple tracking approach such as IoU-based tracking / SORT-style tracking can be considered when video or sequential frames are available.

The purpose is to maintain the identity/location of detected objects across consecutive frames.